

Abdominal/Pelvic CT Angiography (CTA)

ACG: A-0475 (AC)

[Link to Codes](#)

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- Clinical Indications
- Evidence Summary
 - Background
 - Criteria
 - Rationale
 - Related CMS Coverage Guidance
- References
- Footnotes
- Codes

Clinical Indications

- Abdominal/pelvic CT angiography (CTA) may be indicated for **1 or more** of the following(1):
 - Acute limb (ie, lower extremity) ischemia, suspected(4)(5)(6)[N](#)
 - ☐ Aortic aneurysm or dissection evaluation, as indicated by **1 or more** of the following(7)(8)(9)(10)(11):[N](#)
 - Aortic dissection diagnosis and evaluation needed(20)(21)(22)
 - Evaluation of suspected abdominal aortic aneurysm (eg, pulsatile abdominal mass)(23)
 - Intramural hematoma, known or suspected(24)
 - Penetrating atherosclerotic ulcer, known or suspected(24)
 - Postoperative surveillance after surgical or endovascular repair(25)(26)(27)(28)(29)(30)
 - Preoperative evaluation in patient who is candidate for surgical or endovascular intervention(27)(28)(29)(31)
 - Surveillance of unrepaired aortic aneurysm or dissection(28)(29)(32)
 - Vascular connective tissue disorder, known (eg, Marfan syndrome, Loeys-Dietz syndrome, vascular Ehlers-Danlos syndrome), and need for baseline evaluation or regular surveillance based on specific condition(19)(33)
 - ☐ Aortitis or large vessel vasculitis, known or suspected, as indicated by **1 or more** of the following(3)(34)(35)(36):[N](#)
 - Infectious vasculitis (eg, syphilis)(38)
 - Inflammatory vasculitis (eg, Takayasu arteritis, giant cell arteritis, systemic lupus erythematosus)(11)(39)(40)
 - Breast reconstruction (free flap), and need for preoperative planning(41)(42)(43)(44)[N](#)
 - Gastrointestinal bleeding (active)(46)(47)(48)(49)(50)(51)(52)[N](#)
 - ☐ Liver disorder evaluation needed, as indicated by **1 or more** of the following(60)(61):[N](#)
 - Altered or obstructed arterial, venous, or portal flow, suspected(64)
 - Preoperative planning for correction of portal hypertension (eg, placement of transjugular intrahepatic portosystemic shunt)(64)(65)
 - Preoperative planning for liver resection or transplant(62)(63)(66)
 - Preoperative planning for repair of hepatic artery aneurysm
 - Preoperative planning for repair of hepatic venous thrombosis (ie, Budd-Chiari syndrome)
 - Preprocedural planning for catheter placement prior to hepatic artery infusion therapy(67)
 - Vascular complications following liver surgery, suspected(68)(69)
 - ☐ Mass or tumor (abdominal or pelvic), as indicated by **ALL** of the following(70)(71)(72):[N](#)
 - Arterial blood supply evaluation needed
 - Patient candidate for therapeutic intervention (eg, surgery, endovascular treatment, ablation)
 - Tumor of liver, pancreas, stomach, kidney, spleen, or retroperitoneum(76)

- Medium vessel vasculitis, suspected (eg, vasculitis affecting main visceral arteries)(36)(77)[N](#)
- ☐ Mesenteric ischemia (acute or chronic) evaluation needed, as indicated by **1 or more** of the following(78)(79)(80)(81)(82)(83):
[N](#)
 - Acute mesenteric ischemia signs or symptoms (eg, abdominal pain out of proportion to physical examination)(89)(90)
 - Chronic mesenteric ischemia signs or symptoms (eg, postprandial abdominal pain, weight loss, food avoidance)(2)(91)(92)
 - Follow-up of stent placement in mesenteric vessels
- Perineal reconstruction (free flap), and need for preoperative planning(93)[N](#)
- ☐ Peripheral arterial disease, as indicated by **ALL** of the following(3)(94)(95)(96)(97):[N](#)
 - Ankle-brachial index less than or equal to 0.9, or other noninvasive documentation of obstructive occlusive disease[A](101)(102)
 - Arterial disease signs or symptoms, as indicated by **1 or more** of the following(103):
 - Claudication symptoms, chronic (eg, pain; muscle cramping; fatigue; aching or numbness of leg muscles, buttock, or hip associated with walking at reproducible distance and relieved by rest within 10 minutes)
 - Nonhealing lower extremity wound
 - Rest pain or numbness of foot on elevation, relieved by dependency
 - Ulcer or gangrene on lower extremity
 - Revascularization being considered (eg, surgery, angioplasty)(104)
- Preprocedural planning needed prior to transcatheter aortic valve implantation or replacement(105)(106)(107)[N](#)
- ☐ Renal artery pathology, known or suspected, as indicated by **1 or more** of the following(7)(109)(110)(111):[N](#)
 - Fibromuscular dysplasia(114)
 - Postoperative assessment of angioplasty, embolization, or stent placement
 - Renal artery aneurysm or pseudoaneurysm(115)
 - Renal artery arteriovenous fistula
 - Renal artery dissection
 - Renal artery embolism
 - Renal artery stenosis, suspected, as indicated by **1 or more** of the following(116)(117)(118):
 - Accelerated, resistant, or malignant hypertension[B]
 - Atrophic kidney (unexplained), or size discrepancy between the 2 kidneys of greater than 1.5 cm
 - Azotemia or worsening renal function after administration of ACE inhibitor or angiotensin receptor blocker
 - Onset of hypertension in patient younger than 30 years(119)
 - Onset of hypertension in patient older than 55 years
 - Pulmonary edema (sudden and unexplained)
 - Renal failure (unexplained)
 - Renal artery thrombosis(120)
 - Retroperitoneal fibrosis
 - Vasculitis (eg, Takayasu arteritis, polyarteritis nodosa, radiation arteritis)(121)
- ☐ Renal transplant evaluation needed, as indicated by **1 or more** of the following(122):[N](#)
 - Postoperative evaluation of transplant recipient for suspected rejection(125)(126)
 - Postoperative evaluation of transplant recipient for suspected vascular complication (eg, renal artery thrombosis, renal artery stenosis, pseudoaneurysm, arteriovenous fistula) and renal ultrasound inconclusive(127)
 - Preoperative evaluation of graft sites of potential recipient
 - Preoperative evaluation of vascular anatomy of potential donor(128)
- Renal vein thrombosis, suspected(129)[N](#)
- ☐ Renovascular hypertension, suspected, as indicated by **ALL** of the following(110)(130)(131)(132):[N](#)
 - Duplex ultrasonography results indeterminate (eg, due to bowel gas or obesity)(138)
 - Patient is candidate for surgical or endovascular intervention.
 - Strong clinical suspicion of renovascular hypertension (eg, renal bruit, resistant hypertension, known fibromuscular dysplasia, sudden onset of severe hypertension, onset of hypertension in patient younger than 30 years, sudden unexplained pulmonary edema)(139)
- Retroperitoneal bleed, suspected(140)[N](#)
- Splenic artery aneurysm or pseudoaneurysm evaluation needed(3)(74)(115)[N](#)
- Trauma of abdomen or pelvis, known or suspected(3)(11)(141)(142)(143)(144)(145)[N](#)
- Vascular graft complication, suspected (eg, infection, thrombosis, pseudoaneurysm)(11)(38)[N](#)
- ☐ Repeat evaluation of specific area or structure with same imaging modality, and **ALL** of the following:
 - Clinical need for repeat imaging, as indicated by **1 or more** of the following:
 - Change in clinical status (eg, worsening symptoms or new associated symptoms), and findings may impact treatment
 - Need for interval reassessment that may impact treatment plan
 - Need for re-imaging either prior to or after performance of invasive procedure
 - Prior imaging results of specific area or structure with same imaging modality documented and available for comparison

Evidence Summary

Background

CTA consists of CT scanning after intravenous injection of contrast material to better characterize vascular anatomy and pathology. Unlike standard invasive x-ray and digital subtraction angiography, it is noninvasive and rapidly provides information about extravascular and perivascular anatomy. CTA multislice helical scanning with 3-dimensional reconstruction software can provide detailed, visually familiar views of complex anatomy. However, the procedure involves exposure to radiation and contrast material, with risk of allergic reaction and nephrotoxicity.(2)(3) **(EG 2)**

Criteria

The evidence for the clinical indications found in this guideline includes 113 published peer reviewed articles and 26 specialty society or other evidence-based guidelines.

For acute limb (ie, lower extremity) ischemia, suspected, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** A specialty society states that CTA of the abdomen and pelvis with bilateral lower extremity runoff is usually appropriate in the evaluation of an acutely cold painful lower extremity when vascular compromise is suspected.(4) **(EG 2)**

For aortic aneurysm or dissection evaluation, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** CTA has become the preferred technique for assessment of anatomy and determination of patient appropriateness prior to insertion of endovascular aortic stent graft, as well as for surveillance of patients for postprocedure complications, such as endoleak.(8)(12)(13)(14)(15) **(EG 2)** A prospective study of patients who had undergone endovascular abdominal aortic aneurysm repair determined that CTA had sensitivity of 83% and specificity of 100% for detection of endoleaks.(16) **(EG 2)** It is also useful for diagnosing the presence and extent of aortic dissection.(17) **(EG 2)** A study of 62 patients followed after endovascular repair of aortic aneurysm suggested that regular CTA surveillance may be unnecessary, as the combination of Doppler or contrast-enhanced ultrasound and plain x-ray seemed comparable in effectiveness to CTA in follow-up. However, the authors indicated that the decision not to perform CTA should be individualized, with most consideration given to those with unfavorable anatomic characteristics.(18) **(EG 2)** A review article notes that patients with vascular connective tissue disorders, such as Marfan syndrome, Loeys-Dietz syndrome, and vascular Ehlers-Danlos syndrome, are at risk for aneurysms, dissections, and ruptures of medium-sized abdominal arteries and should undergo CTA or MRA of the abdomen and pelvis to evaluate the systemic vasculature at baseline.(19) **(EG 2)**

For aortitis or large vessel vasculitis, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** CTA and MRA have largely replaced conventional angiography for diagnostic confirmation and assessment of infectious or inflammatory vasculitis, as well as post-treatment surveillance.(34)(35)(37) **(EG 2)** CTA and MRA are noninvasive and also reveal information about the vessel wall, surrounding anatomy, and lumen.(35) **(EG 2)**

For breast reconstruction (free flap), and need for preoperative planning, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** Review articles note that CTA may be used to evaluate epigastric and gluteal perforating arteries prior to free flap breast reconstruction. Studies have shown that presurgical CTA may reduce the risk of flap and donor site complications and may also reduce operating time.(42)(43)(45) **(EG 2)** A systematic review and meta-analysis of 14 studies comparing clinical outcomes for preoperative perforator mapping for deep inferior epigastric perforator flap found that there was low-quality evidence that, as compared with no perforator mapping, the use of CTA and MRA was associated with decreased operative time, and as compared with ultrasound, CTA was associated with decreased operative time and lower risk of partial flap failure.(44) **(EG 1)**

For gastrointestinal bleeding (active), evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** Multidetector CTA may be used to evaluate active lower gastrointestinal tract bleeding, including rectal bleeding, particularly when colonoscopy is nondiagnostic.(53) **(EG 2)** A specialty society guideline states that CTA is usually appropriate in the evaluation of active lower gastrointestinal tract bleeding in both hemodynamically stable and hemodynamically unstable patients.(48) **(EG 2)** Specialty society guidelines state that conventional angiography should be performed emergently for hemodynamically unstable patients with active small bowel bleeding. CTA is recommended to evaluate hemodynamically stable patients with evidence of active bleeding, both to identify the site of bleeding and to guide further management. CTA has been shown to detect bleeding rates as slow as 0.3 mL/min, as compared with 0.5 to 1.0 mL/min for conventional angiography.(46)(54) **(EG 2)** A specialty society practice guideline supports the use of CTA as an alternative to visceral angiography when endoscopy confirms the presence of nonvariceal upper gastrointestinal bleeding with or without a clear source, or when there is the clinical presentation of bleeding but the source and presence of blood are not confirmed by endoscopy.(47) **(EG 2)** When endoscopy fails to reveal a diagnosis and continued active bleeding is suspected, CTA may be an appropriate option when standard invasive angiography is either not available or requires an inappropriate delay.(55)(56)(57) **(EG 2)** A systematic review and meta-analysis of 22 studies found that CTA has a sensitivity of 85% and specificity of 92% for the detection and localization of acute active gastrointestinal bleeding.(58) **(EG 1)** A review article notes that CTA can be useful for the evaluation of pancreatic arteriovenous malformation as a cause of gastrointestinal bleeding.(59) **(EG 2)**

For liver disorder evaluation, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** CTA is helpful for delineation of vascular anatomy in the arterial,

venous, and portal circulation, including assessment of patients with cirrhosis, thrombosis, arteriovenous malformations, or aneurysms. (61) **(EG 2)** For liver transplant, CTA has shown effectiveness in evaluation of potential recipients as well as living related donors for adult-adult and adult-child transplants. In addition, it can be used to assess postoperative complications, such as stenosis or thrombosis of anastomosed vessels. (62)(63) **(EG 2)**

For mass or tumor of the abdomen or pelvis, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** CTA can produce detailed and accurate 3-dimensional maps of vessels when planning surgery or ablation of masses in the liver, pancreas, stomach, spleen, kidney, or retroperitoneum. (71)(73)(74) **(EG 2)** A prospective study of 89 patients with pancreatic cancer, which included 60 patients randomized to undergo CTA, found that the rate of visualization of the great vessels around the pancreas was 100% with CTA; the authors concluded that peripancreatic vascular reconstruction by CTA can reveal vascular anatomy and changes associated with pancreatic tumors, thereby impacting surgical management. (75) **(EG 2)**

For medium vessel vasculitis, suspected, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** A specialty society guideline states that CTA is usually appropriate in the evaluation of suspected medium vessel vasculitis. (36) **(EG 2)**

For mesenteric ischemia (acute and chronic), evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** Acute mesenteric ischemia occurs secondary to acute embolism of the superior mesenteric artery, thrombosis of the superior mesenteric artery, nonocclusive mesenteric ischemia, or mesenteric and portal vein thrombosis. Chronic mesenteric ischemia is most commonly caused by severe atherosclerotic vascular disease, with rare causes including fibromuscular dysplasia, median arcuate ligament syndrome, dissection, or vasculitis. (78) **(EG 2)** Specialty society guidelines support the use of CTA for the diagnosis of both acute and chronic mesenteric ischemia, with reported sensitivities of 93% and 96%, respectively, and specificities of 100% and 94%, respectively. (78)(79)(83) **(EG 2)** For the diagnosis of acute mesenteric ischemia, specialty society guidelines support the use of CTA as the preferred imaging study; although MRA has a high sensitivity and specificity for diagnosing stenosis at the origins of the celiac axis and superior mesenteric artery, it has a limited role in diagnosing distal stenosis or nonocclusive mesenteric ischemia. (78)(83) **(EG 2)** For the diagnosis of chronic mesenteric ischemia, specialty society guidelines state that CTA is considered the imaging method of choice. MRA has sensitivity and specificity of 100% and 95%, respectively, for the diagnosis of chronic mesenteric ischemia and may be considered as an alternative to CTA. However, the inferior mesenteric artery and peripheral mesenteric vessels are better visualized with CTA. (78)(83) **(EG 2)** CTA can be used to evaluate the presence and degree of stenosis or dissection (84)(85) in major vessels, as well as the extent of collateral circulation. (80)(86) **(EG 2)** It is also useful in following stent placement or vascular anastomoses for abdominal angina, without significant artifact from clips or stent that is often seen with MRA. (87) **(EG 2)** A review article notes that newer 3-dimensional multidetector CT scanning techniques allow for greater visualization of splanchnic vessels than do standard axial CT scan images. (88) **(EG 2)**

For perineal reconstruction (free flap), and need for preoperative planning, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** A review article notes that CTA can be useful for preoperative planning of perineal flap reconstruction following radical pelvic surgery. CTA can visualize the deep inferior epigastric arteries, which may influence decisions regarding perineal reconstruction. (93) **(EG 2)**

For peripheral arterial disease, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** A systematic review of 107 studies concluded that duplex (Doppler) ultrasound scan, CTA, and MRA have similar accuracy for the detection of an occlusion or a stenosis of 50% or more above and below the knee. (98) **(EG 1)** Studies using multislice CTA have shown near-perfect sensitivity and specificity, comparable with those of digital subtraction angiography. However, the presence of vessel wall calcifications can decrease the utility of CTA. MRA compares favorably to digital subtraction angiography in determination of the presence of stenoses, and it has the advantage of avoiding arterial puncture and potential nephrotoxicity from iodine-based contrast material. In most cases, CTA and MRA are not necessary for the initial diagnosis of peripheral arterial disease, which is typically confirmed through clinical evaluation and noninvasive studies, including the ankle-brachial index. (87)(94)(99) **(EG 2)** Specialty society guidelines note that CTA, MRA, and duplex (Doppler) ultrasound are all useful to diagnose the anatomic location of peripheral arterial disease, as well as the degree of stenosis, in symptomatic patients in whom revascularization is being considered. (94)(97) **(EG 2)**

For preprocedural planning needed before transcatheter aortic valve implantation or replacement, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** Subspecialty practice guidelines recommend CT angiogram of the chest, abdomen, and pelvis for assessment of access vessels (femoral, subclavian, iliac, aorta) before proceeding with transcatheter aortic valve implantation or replacement. (105)(106)(107)(108) **(EG 2)**

For renal artery pathology, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** A review article considers CTA and MRA to be appropriate imaging modalities, with CTA providing better spatial resolution and reduced examination time and MRA being preferable when it is desirable to avoid radiation exposure or iodinated contrast. (110)(112) **(EG 2)** CTA has mean sensitivity and specificity of 94% and 93%, respectively, for the detection of hemodynamically significant atherosclerotic renal artery stenosis. (113) **(EG 2)** An evidence-based guideline endorsed by multiple specialty societies states that CTA is recommended in patients with normal renal function as a screening test to establish the diagnosis of renal artery stenosis. (7) **(EG 2)**

For renal transplant, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** CTA can be used for pretransplant imaging of both donor and recipient; it also can be used for assisting with appropriate surgical planning, including better awareness of vessel anatomy during laparoscopic donation, and of appropriate placement in recipients, especially older patients, for whom understanding of the location of atherosclerotic plaque, calcification, and stenosis is helpful.(122) **(EG 2)** A study comparing CTA with MRA for preoperative assessment of living kidney donors found comparable, extremely high sensitivity for assessment of renal arteries (100%) and veins (97%); both imaging modalities were judged to be appropriate diagnostic options, with selection dependent on individual potential donor preferences and clinical requirements.(123)(124) **(EG 2)**

For renal vein thrombosis, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** CTA has a sensitivity and specificity approaching 100% and is thought by many to be the imaging modality of first choice. MRA is appropriate when CTA is unavailable or contraindicated.(129) **(EG 2)**

For renovascular hypertension, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** Review articles of renal MRA and CTA state that both imaging modalities are highly accurate for the detection of significant renal artery stenosis and can be used safely as noninvasive alternatives to conventional diagnostic angiography.(110)(133)(134) **(EG 2)** A guideline endorsed by multiple specialty societies states that duplex ultrasonography, CTA (in patients with normal renal function), and MRA may all be used as screening tests to establish the diagnosis of renal artery stenosis.(7) **(EG 2)** In several single-center series, both CTA and MRA, compared with digital subtraction angiography, had sensitivity over 90% and specificity over 80%, with little difference between abdominal CTA and MRA.(135)(136) **(EG 2)** A prospective study of 78 patients with hypertension and suspected renal artery stenosis determined that contrast-enhanced MRA had sensitivity of 90% and specificity of 96% for detection of stenosis greater than 60%.(137) **(EG 2)** However, each modality presents unique technical problems: vascular calcification decreases the ability to see the arterial lumen or grade the stenosis with CTA, making MRA preferable in elderly patients; and the presence of a stent can create image artifact on MRA that makes diagnosis of in-stent stenosis impossible, making CTA preferable in this clinical situation.(87) **(EG 2)**

For retroperitoneal bleed, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** A specialty society guideline states that CTA is usually appropriate when evaluating suspected retroperitoneal bleeding.(140) **(EG 2)**

For splenic artery aneurysm or pseudoaneurysm evaluation, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** A specialty society practice guideline notes that CTA can be useful for the evaluation of splenic artery aneurysms.(3) **(EG 2)** A review article notes that multidetector-row CTA, with its high spatial and temporal resolution and postprocessing tools, allows for detailed visualization of the splenic vascular anatomy, including potential aneurysms and pseudoaneurysms that may affect the splenic artery.(74) **(EG 2)**

For trauma of the abdomen or pelvis, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** CTA is a quick and accurate means of assessing patients; it provides information about vascular integrity as well as the usual comprehensive structural and anatomic information provided by standard CT scan.(3)(141)(146) **(EG 2)** Sensitivity of 93% to 94% has been reported for detection of active bleeding, including differentiation between active arterial and active venous hemorrhage, which can help predict the need for surgery.(147) **(EG 2)** A systematic review and meta-analysis of 11 studies (891 trauma patients) found that CTA has sensitivity of 96% and specificity of 99% for the diagnosis of trauma-related arterial injury.(148) **(EG 2)** Studies of trauma-related solid organ and pelvic bleeding report embolization mean clinical success rates of 91% for kidney, 85% for spleen, 80% for liver, and 92% for pelvic injury.(149)(150) **(EG 2)**

For vascular graft complication, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** A review article notes that CTA can be used in the evaluation of vascular graft infections.(38) **(EG 2)** A retrospective study of 20 patients who had clinical signs and symptoms of vascular graft infection showed that CTA had sensitivity of 88% and specificity of 50%.(151) **(EG 2)** A specialty society practice guideline supports the use of CTA to assess the patency and integrity of vascular graft reconstruction.(3) **(EG 2)**

Rationale

Use of this MCG care guideline helps the clinician determine if a particular treatment, medication, or service might be appropriate for a specific patient, taking into account their unique health complexities.

Use of these evidence-based clinical criteria to support decision making benefits the patient by identifying patient-specific complex clinical factors and conditions, promoting personalized treatment. Utilizing evidence-based clinical criteria promotes patient safety by helping ensure that potential patient benefits outweigh the risks. In addition, the use of evidence-based guidelines can increase consistency in treatment thresholds, leading to less variation in care and promoting equitable treatment among patients.

Related CMS Coverage Guidance

This guideline supplements but does not replace, modify, or supersede existing Medicare regulations or applicable National Coverage Determinations (NCDs) or Local Coverage Determinations (LCDs).

Code of Federal Regulations (CFR): 42 CFR 419.22(152); 42 CFR 422.101(153)

Internet-Only Manual (IOM) Citations: CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 14 - Medical Devices(154); CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 15 - Covered Medical and Other Health Services(155); CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 16 - General Exclusions from Coverage(156)

Medicare Coverage Determinations: Medicare Coverage Database(157)

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Footnotes

[A] Patients with suspected lower extremity peripheral arterial disease should undergo measurement of the ankle-brachial index. A normal ankle-brachial index value is 1.0 to 1.4; a value greater than 1.4 indicates noncompressible arteries, a value of 0.91 to 0.99 is considered borderline, and a value less than or equal to 0.90 is abnormal and indicates vascular occlusive disease of the lower extremity. A postexercise measurement should be made in patients if the resting ankle-brachial index is normal and the results would change management.(94)(96)(100) [A in Context Link 1]

[B] Accelerated hypertension is defined by sudden and persistent worsening of previously controlled hypertension. Resistant hypertension is defined as the failure to achieve blood pressure goals in patients who are adherent to an appropriate 3-drug antihypertensive regimen. Malignant hypertension is hypertension with evidence of acute end organ damage, such as decompensated heart failure, renal failure, or retinopathy.(7) [B in Context Link 1]

Codes

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